Probabilistic methods in Combinatorics

Submitted by -

Pallav Goyal
4th year Undergraduate Student
Department of Mathematics and Statistics
Indian Institute of Technology Kanpur

Mentored by -

Dr. Somnath Jha
Assistant Professor
Department of Mathematics and Statistics
Indian Institute of Technology Kanpur

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1 Introduction

Combinatorics is a branch of mathematics that deals with counting and studying properties of discrete structures, generally finite or countable. Over the years, combinatorics has generated significant interest among mathematicians because of its exoteric nature, the plethora of problems available to be tackled and the diversity of techniques from other fields of mathematics that can be used for solving combinatorial problems. We'd like to briefly discuss one such technique in this report.

Probabilistic methods for solving mathematical problems were popularised by Paul Erdös. The general idea involved is to use non-determinism to show that certain statements are true with positive probability over randomly selected configurations, thereby proving the existence of at least one such configuration where the the statement is true. Although this is (almost always) a non-constructive method, the applications are wide-ranging, as we'll see, from problems in additive combinatorics to problems in graph theory.

The organisation of the report is as follows. In order not to disrupt the flow of theory and problems and keep them independent from each other, we divide the report into 2 sections. Section 2 covers all the theory and techniques that we'll be needing over the course of our study. Section 3 uses the techniques developed in section 2 for solving combinatorial problems. Most of the pre-requisites have been discussed and can be read up from the references. Almost the entire content can be referred to from [1]. It is assumed that the reader has a good command over rigorous proofs and arguments, elementary number theory, some basic probability theory, graph theory and a familiarity with the Big-Oh notation. Logarithms will be taken with base 2 unless explicitly mentioned.

2 Theory

Let S be an underlying sample space. We use E to denote an event in S. We denote by $P(E)$ and $I(E)$ respectively the probability and the indicator function of E. Apart from this, we use standard Boolean algebra notation for unions, complements and intersections. As we'll be dealing exclusively with discrete random variables, we define the expectation $E[X]$ of a random variable X as:

$$E[X] := \sum_x xP(X = x)$$

and variance $Var(X)$ as:

$$Var(X) := E[X^2] - E[X]^2$$

For 2 random variables X, Y, we define their covariance $Cov(X, Y) = E[XY] - E[X]E[Y]$.

One notes that for a given event A, $E[I(A)] = P(A)$. Now, for a given event Y such that $P(Y) > 0$, we define for any event X the conditional probability of X given Y as:

$$P(X|Y) = \frac{P(X \wedge Y)}{P(Y)}$$

The simplest and the most fundamental result that we'll be using is the following inequality that helps give bounds on probability by means of expectation.

Theorem 2.1. *(Markov's Inequality) Given a non-negative random variable X, for any positive real λ,*

$$P(X \geq \lambda) \leq \frac{E[X]}{\lambda}$$

Proof. Consider the inequality $X \geq \lambda I(X \geq \lambda)$ which is true as $X \geq 0$. Then, taking expectations on both sides and dividing by λ, we get the desired inequality. □

What this tells us is that the probabilty of X being much larger than its expectation is very low. A simple application of this inequality gives the following deep result.

Lemma 2.2. *(Borel-Cantelli Lemma) Let $E_1, E_2, E_3, \cdots$ be a sequence of events such that $S = \sum_{n=1}^{\infty} P(E_n) < \infty$. Then, for any positive integer M, we have that P(Fewer than M events from E_1, E_2, E_3 . . . hold) $\geq 1 - \frac{S}{M}$*

Proof. Let Y_n denote the number of events out of $E_1, E_2, \ldots E_n$ which hold. Then, $E[Y_n] = \sum_{k=1}^{n} P(E_k)$. Then, by Markov's inequality,

$$P(Y_n \geq M) \leq \frac{\sum_{k=1}^{n} P(E_k)}{M}.$$

In other words,

$$P(Y_n \leq M) \geq 1 - \frac{\sum_{k=1}^{n} P(E_k)}{M}$$

, and so, taking the limit as n goes to infinity, we complete the proof. (Here, we have used the continuity of the probability function when we take the limit.) □

Although Markov's inequality gives us a useful bound, it has several limitations. For example, it doesn't say anything about a lower bound on the values taken by X. The reason for this is that it doesn't use any information other than $E[X]$. Also using $Var(X)$, we get the following result:

Theorem 2.3. *(Chebyshev's inequality) For any random variable X and positive real λ,*

$$P(|X - E[X]| > \lambda Var(X)^{1/2}) \leq \frac{1}{\lambda^2}$$

Proof. The inequality trivially holds when $Var(X) = 0$, so let $Var(X) > 0$. Noting that $E[(X - E[X])^2] = Var(X)$, we have by Markov's inequality:

$$P(|X - E[X]|^2 \geq \lambda^2 Var(X)) \leq \frac{Var(X)}{\lambda^2 Var(X)} = \frac{1}{\lambda^2}$$

and so, the claim follows. □

This basically says that X takes the value $E[X] + O(Var(X)^{1/2})$ with probability $1 - O(\lambda^{-2})$. A further improvement to this result for large λ is provided by Chernoff's inequality. To prove that we need a lemma:

Lemma 2.4. *If X is a random variable with $|X| \leq 1$ and $E[X] = 0$, for any $-1 \leq t \leq 1$, we have $E[e^{tX}] \leq e^{t^2 Var(X)}$*

Proof. As $|tX| \leq 1$, using elementary differential calculus and Taylor series, we have the inequality:

$$e^{tX} \leq 1 + tX + t^2X^2$$

Then, taking expectations on both sides, we get the inequality as:

$$E[e^{tX}] \leq 1 + E[t^2X^2] = 1 + t^2Var(X) \leq e^{t^2Var(X)}.$$

□

Theorem 2.5. *(Chernoff's inequality) Given independent random variables $X_1, X_2, \ldots X_n$ such that $|X_i - E[X_i]| \leq 1$ for all i, if $X = X_1 + X_2 + \cdots + X_n$ such that $\sigma^2 = Var(X)$, then for any $\lambda > 0$,*

$$P(|X - E[X]| \geq \lambda\sigma) \leq 2e^{-\lambda^2/4}$$

Proof. Without loss of generality, $E[X_i] = 0$ for all i. We have

$$P(X \geq \lambda\sigma) = P(tX \geq t\lambda\sigma) = P(e^{tX} \geq e^{t\lambda\sigma}) = \frac{E[e^{tX}]}{e^{t\lambda\sigma}}$$

by Markov's inequality, where we have assumed that $t > 0$. By independence, we have

$$E[e^{tX}] = E[e^{tX_1}]E[e^{tX_2}] \cdots E[e^{tX_n}].$$

By the previous lemma, $E[e^{tX_i}] \leq e^{t^2Var(X_i)}$. Now, as the X_i's are independent, $Cov(X_i, X_j) = 0$ whenever $i \neq j$, and so, $Var(X) = Var(X_1) + Var(X_2) + \cdots + Var(X_n)$. Putting all the pieces together,

$$P(X \geq \lambda\sigma) \leq e^{-t\lambda\sigma}e^{t^2\sigma^2}.$$

Symmetrically, we can estimate $P(X \leq -\lambda\sigma)$. Cumulatively, we get:

$$P(|X| \geq \lambda\sigma) \leq 2e^{-t\lambda\sigma}e^{t^2\sigma^2}$$

and so, putting $t = \frac{\lambda}{2\sigma}$, we get our inequality. □

Corollary 2.6. *Given independent Boolean random variables $t_1, t_2, \ldots, t_n$ and X, the sum of these variables, for any $\epsilon > 0$, we have:*

$$P(|X - E[X]| \geq \epsilon E[X]) \leq 2e^{-\frac{\epsilon^2 E[X]}{4}}$$

Proof. The proof follows directly from the previous theorem taking $\lambda = \frac{\epsilon E[X]}{\sigma}$ and observing that expectation upper bounds variance for Boolean random variables, implying $E[X] \geq Var(X)$ due to independence. □

Next, we talk about an inequality that relates monotone random variables. If $t_1, t_2, \ldots t_n$ are independent Boolean random variables, a random variable $X = X(t_1, t_2, \ldots t_n)$ is said to be monotone if $X(t'_1, t'_2, \ldots t'_n) \geq X(t_1, t_2, \ldots t_n)$ whenever $t'_i \geq t_i$ for all i. An event E is the sample space is said to be monotone increasing (respectively decreasing) if the random variable $I(E)$ is monotone increasing (respectively decreasing).

Theorem 2.7. *(FKG inequality) Let X and Y be both monotone increasing (or both monotone decreasing) random variables. Then,*

$$E[XY] \geq E[X]E[Y]$$

Proof. Without loss of generality, we assume that X and Y are monotone increasing. We use induction on n, where n denotes the number of Boolean random variables X and Y depend on, say $t_1, t_2, \ldots, t_n$. The case $n = 0$ is trivially true.

Now, assume that $P(t_n = 1)$ and $P(t_n = 0)$ are both non-zero, because otherwise, we are deterministically in the case of $n - 1$ Boolean variables where the result is true by the induction hypothesis. By shifting if needed, we can also assume that $E[X|t_n = 0] = E[Y|t_n = 0] = 0$. Then, by monotonicity, we have

$$E[X|t_n = 1]E[Y|t_n = 1] \geq 0.$$

Also, by induction hypothesis, $E[XY|t_n = 0] \geq E[X|t_n = 0]E[Y|t_n = 0] = 0$ and $E[XY|t_n = 1] \geq E[X|t_n = 1]E[Y|t_n = 1]$. Then, we have

$$\begin{aligned} E[XY] &= E[XY|t_n = 0]P(t_n = 0) + E[XY|t_n = 1]P(t_n = 1) \\ &\geq E[X|t_n = 1]E[Y|t_n = 1]P(t_n = 1) \\ &\geq E[X|t_n = 1]P(t_n = 1)E[Y|t_n = 1]P(t_n = 1) \\ &= E[X]E[Y] \end{aligned}$$

□

Corollary 2.8. *If $A_1, A_2, \ldots A_n$ are increasing events, then*

$$P(A_1 \wedge A_2 \wedge \cdots \wedge A_n) \geq P(A_1)P(A_2)\cdots P(A_n)$$

Our final result in this section is going to be a result that shows that if we have a dependency graph for a finite set of events, then, with a positive probability, none of the events occur provided some relations are true regarding the individual and conditional probabilities.

Theorem 2.9. *(Lovász local lemma) For a finite set V, suppose we have an event A_i corresponding to each $i \in V$. Suppose we have a directed graph $G(V, E)$ without any loops and a sequence of real numbers $0 \leq x_i < 1$ for each $i \in V$ such that*

$$P(A_i | \bigwedge_{j \in S} \overline{A_j}) \leq x_i \prod_{(i,j) \in E} (1 - x_j)$$

where for a fixed $i \in V$, $S \subseteq V \setminus \{i\}$ such that the above probability is finite and $(i, j) \notin E$ for any $j \in S$. Then, for disjoint $S, S' \subseteq V$, we have:

$$P(\bigwedge_{i \in S} \overline{A_i} | \bigwedge_{j \in S'} \overline{A_j}) \geq \prod_{i \in S} (1 - x_i) > 0$$

Proof. The proof is by induction on $|S| + |S'|$. The theorem is trivially true when $|S| = 0$ and so, we assume that $|S| \geq 1$. Now, if $|S| > 1$, let $S = \{j\} \cup S \setminus \{j\}$ for some j. Then, by the definition of conditional probabilty,

$$P(\bigwedge_{i \in S} \overline{A_i} | \bigwedge_{i \in S'} \overline{A_i}) = P(\overline{A_j} | \bigwedge_{i \in S' \cup S \setminus \{j\}} \overline{A_i}) P(\bigwedge_{i \in S \setminus \{j\}} \overline{A_i} | \bigwedge_{i \in S'} \overline{A_i})$$

and we can use the induction hypothesis on the second term. Thus, we may assume that $|S| = 1$.

Let $S = \{i\}$ and $S' = S_1 \cup S_2$ where $S_1 = \{j \in S' : (i, j) \in E\}$ and $S_2 = S' \setminus S_1$. Then, we have

$$P(A_i | \bigwedge_{j \in S'} \overline{A_j}) = \frac{P(A_i, \bigwedge_{j \in S_1} \overline{A_j} | \bigwedge_{j \in S_2} \overline{A_j})}{P(\bigwedge_{j \in S_1} \overline{A_j} | \bigwedge_{j \in S_2} \overline{A_j})}.$$

For the numerator, using the hypothesis of the theorem, we notice that:

$$P(A_i, \bigwedge_{j\in S_1} \overline{A_j} | \bigwedge_{j\in S_2} \overline{A_j}) \leq P(A_i | \bigwedge_{j\in S_2} \overline{A_j}) \leq x_i \prod_{j\in V|(i,j)\in E} (1-x_j)$$

and for the denominator, as $|S_1|+|S_2| < 1+|S|$, by the induction hypothesis:

$$P(\bigwedge_{j\in S_1} \overline{A_j} | \bigwedge_{j\in S_2} \overline{A_j}) \geq \prod_{j\in S_1}(1-x_j) \geq \prod_{j\in V|(i,j)\in E} (1-x_j).$$

Then, by dividing the above 2 inequalities, we get the desired result. □

Corollary 2.10. *Let $d \geq 1$ and $0 < p < 1$ such that:*

$$p \leq \frac{1}{e(d+1).}$$

Let V be a finite set such that for all $i \in V$, A_i is a probabilistic event with probability bounded by p. Also, suppose for a fixed i, A_i is mutually independent with all but at most d of the other events. Then,

$$P(\bigwedge_{i\in V} \overline{A_i}) \geq \left(1 - \frac{1}{d+1}\right)^{|V|} > 0$$

Proof. Taking $x_i = \frac{1}{d+1}$ and constructing the dependency graph $G(V, E)$ where $(i, j) \in E$ if and only if A_i and A_j are not pairwise independent, we are done using the previous theorem after observing that $P(A_i) \leq p \leq \frac{1}{e(d+1)} \leq \frac{1}{d+1}\left(1 - \frac{1}{d+1}\right)^d$ (by the definition of the exponential constant e.) □

3 Applications

Now we use the results developed in the previous section to solve combinatorial problems.

3.1 Graph Theory

Given an undirected graph $G = G(V, E)$, we call a subset S of V an independent set if for all $i, j \in S$, $(i, j) \notin E$. For any $v \in V$, we use the notation $N(v)$ to denote the set of all neighbours of v.

Theorem 3.1. *(Turán's Theorem) Let $G = G(V, E)$ be a graph on n vertices. Then, G contains an independent set of size $\sum_{v \in V} \frac{1}{deg(v)+1}$.*

Proof. Let $\pi : V \to \{1, 2, \ldots n\}$ be a bijection picked at random. Call a vertex v *great* if for all $w \in N(v)$, $\pi(v) > \pi(w)$. Then, the set S of all *great* vertices in a graph for a fixed π is an independent set. Also, for a given vertex v, the probability that v is a *great* vertex is equal to the probability that $\pi(v)$ is greater than $\pi(w)$ for all $w \in N(v)$. As π has been picked at random, this probability is equal to $\frac{1}{deg(v)+1}$. So, we have:

$$E[|S|] = \sum_{v \in V} P(v \in S) = \sum_{v \in V} \frac{1}{deg(v) + 1}.$$

Therefore, there exists π such that $|S| \geq \sum_{v \in V} \frac{1}{deg(v)+1}$ and so, we're done. □

The above proof demonstrates that even without the use of any strong theorems or inequalities about probabilities and expectations, we can write short proofs for seemingly non-trivial results. This is a trend we'll see multiple times.

Theorem 3.2. *Let $G = G(V, E)$ be a triangle-free graph on n vertices and at least 1 edge such that the maximum degree of a vertex is d. Then, G has an independent set of size $\Omega(\frac{n}{d} \log d)$.*

Proof. Let X be the set of all independent sets in G. X is non-empty as all the singletons belong to X. We'll be done if we can show that for a randomly

picked I from X, $E[|I|] \geq \frac{n \log d}{8d}$.

For each vertex $v \in V$, we define a random variable:

$$Y_v := d|I \cap \{v\}| + |N(v) \cap I|$$

Since I is independent, for $v \in I$, we have $N(v) \cap I = \phi$, and so, $Y_v = d$. For $v \notin I$, we have $Y_v = |w \in I : (v, w) \in E|$. As each vertex in I can be a neighbour of at most d other vertices, we have the inequality

$$\sum_{v \in V} Y_v \leq 2d|I|$$

, and so, $E[|I|] \geq \frac{1}{2d} \sum_{v \in V} E[Y_v]$. Therefore, it suffices to show that

$$E[Y_v] \geq \frac{\log d}{4}$$

for all $v \in V$.

Fix v. Consider the induced subgraph $G' = G(V', E')$ of G where $V' = V \setminus \{N(v) \cup \{v\}\}$. Then, $I' = I \cap V'$ is an independent set of V'. Fix this I'. We'll try to compute $E[Y_v | I \cap V' = I']$

Let $J \subseteq N(v)$ be the set of those neighbours of v that aren't adjacent to any element of I'. Let $|J| = m$. As the elements of J are neighbours of v, by the triangle-free nature of G, J must be an independent set. Thus, we can conclude that given I', I can be constructed either by adding v or a subset of J to I'. Thus, as there are 2^m subsets of J, we have $2^m + 1$ choices for I altogether. If v is added to I', $Y_v = d$. If a subset $J' \subseteq J$ is added to I', $Y_v = |I'|$. So, we have

$$E[Y_v | I \cap V' = I'] = \frac{d}{2^m + 1} + \frac{m2^{m-1}}{2^m + 1}$$

as $m2^{m-1}$ is the sum of the cardinalities of all the subsets of J'. We claim that the above expression is bounded below by $\frac{\log d}{4}$. The claim is trivially true when $d \leq 1$ or when $m = 1$. For $d > 1$ and $m > 1$, we proceed by

induction on m. Then,

$$\begin{aligned}
\frac{2^{\log d} + (m+1)2^m}{2^{m+1}+1} &= \frac{2^{\log d - 1} + (m+1)2^{m-1}}{2^m + 1/2} \\
&\geq \frac{2^{\log d - 1} + (m+1)2^{m-1}}{2^m + 1} \\
&= \frac{2^{\log d - 1} + m2^{m-1}}{2^m+1} + \frac{2^{m-1}}{2^m+1} \\
&\geq \frac{\log d - 1}{4} + \frac{2}{3} \\
&\geq \frac{\log d}{4}
\end{aligned}$$

Hence, the inequality stands proven. As this is true for all such I', by total probability theorem, we conclude with the same inequality for $E[Y_v]$. This completes the proof of the theorem. □

Now, leaving behind the concept of independent sets, we talk about crossing number of a graph. Embedding a graph in $\mathbb{R}^2$, we define the crossing number $cross(G)$ for a particular embedding of a graph $G = G(V, E)$ to be the number of unordered pairs $(e_1, e_2) \in E^2$ such that e_1 and e_2 intersect, but do not share a vertex. A graph is called a planar graph if $cross(G) = 0$. We'll prove a lower bound on the crossing number for a given graph, but first, we prove a purely graph theoretic lemma.

Lemma 3.3. *For a planar graph $G = G(V, E)$, we must have $|E| \leq 3|V|$.*

Proof. The proof is clear when $|V| = 1, 2$, so let $|V| \geq 3$. By Fáry's Theorem, we can assume that all the edges are straight line segments. Now, for any internal face S formed in the planar graph, it must be bounded by at least 3 edges. Hence, if $|F|$ denotes the total number of faces in G, the number of internal faces is $|F| - 1$, and so, the number of edges E is at least $3(|F| - 1)/2$ (as each edge can be a part of at most 2 faces). Thus,

$$|F| \leq \frac{2}{3}|E| + 1.$$

Using Euler's formula $|E| + 2 = |V| + |F|$ and the above inequality, we get the desired inequality. □

Theorem 3.4. *Given a graph $G = G(V, E)$ such that $|E| \geq 4|V|$, $cross(G) \geq \frac{|E|^3}{64|V|^2}$.*

Proof. Now, as there are $cross(G)$ crossings in G, by removing at most $cross(G)$ edges, we can make G planar. Thus, by the previous lemma, we can conclude that $cross(G) \geq |E| - 3|V|$.

Let $0 < p \leq 1$ and consider randomly constructed $V' \subseteq V$ such that for any vertex $v \in V$, the independent probability of belonging to V' is p. Let $G' = (V', E')$ be the induced subgraph. Then, we have the inequality:

$$E[cross(G')] \geq E[|E'|] - 3E[|V'|] = p^2|E| - 3p|V|$$

Also, as any crossing in G has p^4 probability of surviving in G', we conclude that

$$p^4 cross(G) \geq p^2|E| - 3p|V|$$

Putting $p = 4|V|/|E|$ gives us the required inequality, □

3.2 Additive bases

Given a subset B of $\mathbb{N}$, we denote by $r_{k,B}(n)$ to be the number of ways of expressing n as a sum k elements of B. We say that B is a basis of $\mathbb{N}$ of order k if $r_{k,B}(n) \geq 1$ for all sufficiently large n.

Theorem 3.5. *There exists a basis B of $\mathbb{N}$ of order 2 such that $r_{2,B}(n) = \Theta(\log n)$ for sufficiently large n.*

Proof. Consider the set $B \subseteq \mathbb{N}$ defined randomly such that for any positive integer n, we have the jointly independent probability:

$$P(n \in B) = \left(C\sqrt{\frac{\log n}{n}}, 1\right)$$

where C is a constant that will be decided upon later. We want to show that for a suitably chosen C, $r_{2,B}(n) = \Theta(\log n)$ for sufficiently large n with positive probability (taken over the choice of B). Now,

$$E[r_{2,B}(n)] = \sum_{1 \leq i < n/2} E[I(i \in B)I(n - i \in B)] + O(1)$$

$$= \sum_{1\le i<n/2} C^2\sqrt{\frac{\log i \log(n-i)}{i(n-i)}} + O(1)$$
$$= \Theta\Big(\sqrt{\frac{\log n}{n}} \sum_{1\le i<n/2} \sqrt{\frac{\log i}{i}}\Big) + O(1)$$

Now, $\sum_{1\le i<n/2} \sqrt{\frac{\log i}{i}} \le \sqrt{\log n} \sum_{1\le i<n/2} \frac{1}{\sqrt{i}} = \sqrt{\log n}\; O(\sqrt{n})$, and $\sum_{1\le i<n/2} \sqrt{\frac{\log i}{i}} \ge \frac{1}{\sqrt{n}} \sum_{1\le i<n/2} \sqrt{\log i} = \frac{1}{\sqrt{n}} O(n\sqrt{\log n})$. Thus, we conclude that $E[r_{2,B}(n)] = \Theta(\log n)$. By choosing a large enough C, we can have $32\log n \le E[\sum_{1\le i<n/2} I(i \in B)I(n-i \in B)] \le \kappa \log n$ for some $32 < \kappa$ for suitably large n, and so, by Corollary 2.6 for the case $\epsilon = 1/2$,

$$P(\sum_{1\le i<n/2} I(i \in B)I(n-i \in B) > \frac{3\kappa}{2}\log n) \le \frac{2}{n^2}$$

and

$$P(\sum_{1\le i<n/2} I(i \in B)I(n-i \in B) < 16\log n) \le \frac{2}{n^2}.$$

Therefore, from the above two inequalities, we have that $P(\sum_{1\le i<n/2} I(i \in B)I(n-i \in B) = \Theta(\log n)) = 1 - O\Big(\frac{1}{n^2}\Big)$. Now, as $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, by Lemma 2.2, we conclude that all but finitely many of the events $X_n : \sum_{1\le i<n/2} I(i \in B)I(n-i \in B) = \Theta(\log n)$ happen. Hence, $r_{2,B}(n) = \Theta(\log n)$ for all but finitely many n, and hence, we're done. □

Given a subset A of $\mathbb{N}$, B is called a complementary basis of A of order k if all sufficiently large n can be expressed as a sum of an element of A and k elements of B.

Theorem 3.6. *The set P of all prime numbers has a complementary basis of order 1 such that $|B \cap [1,n]| = O(\log^2 n)$ for sufficiently large n.*

Proof. The strategy to be followed is similar to the previous theorem. We construct a subset B of $\mathbb{N}$ randomly by independently selecting elements such

that:

$$P(n \in B) = min\Big(C\frac{\log n}{n}, 1\Big)$$

for a suitable constant C. Then,

$$E[|B \cap [1,n]|] = C\sum_{k=1}^{n}\frac{\log k}{k} \leq C\log n\sum_{k=1}^{n}\frac{1}{n} = O(\log^2 n).$$

Thus, by an application of Corollary 2.6 and using a similar argument as in the previous theorem, $|B \cap [1,n]| = O(\log^2 n)$ for all sufficiently large n as a consequence of Lemma 2.2. Now, let $r_{P+B}(n)$ denote the number of ways of expressing n as a sum of an element each from P and B. We wish to show now that $r_{P+B}(n) > 0$ for all sufficiently large n with positive probability. Again, by applying Lemma 2.2, we only need to show that:

$$P(r_{P+B}(n) > 0) = 1 - O\Big(\frac{1}{n^2}\Big)$$

for sufficiently large n. We notice that:

$$\begin{aligned} E[r_{P+B}(n)] &= E[|B \cap (n-P|] \\ &= C\sum_{p \in P \cap [1,n]}\frac{\log(n-p)}{n-p} \\ &= \Omega(\log n) \end{aligned}$$

where the final equality follows from a standard estimate in analytic number theory (see Proposition 1.54 in [1]). Thus, for a suitable choice of C (large enough to meet the requirements here as well as for the previous application of Corollary 2.6), $E[B \cap [1,n]] > 32\log n$, and so, another application of Corollary 2.6 does the job. □

For a subset A of $\mathbb{N}$, we define the lower density $\underline{\sigma}(A)$ as:

$$\underline{\sigma}(A) := \liminf_{n\to\infty}\frac{A \cap [1,n]}{n}$$

We call X a $(1-\epsilon)$-complementary basis of A of order k if $\underline{\sigma}(A+kX) \geq 1-\epsilon$.

Theorem 3.7. *For the set P of prime numbers and any $0 < \epsilon < 1$, there exists a $(1-\epsilon)$-complementary basis of P of order 1 such that $|X \cap [1,n]| = O(\log n)$.*

Proof. The following lemma forms the crux of the proof of the theorem:

Lemma 3.8. *For any $\epsilon > 0$ and for all large enough n depending on ϵ, there exists a set $B \subseteq [n^{2/3}, 2n^{2/3}]$ with $|B| = O(\log n)$ and*

$$|[1, x] \setminus (P + B)| \leq \epsilon x$$

for all $n^{3/4} \leq x \leq n$.

Proof. Choose $B \subseteq [n^{2/3}, 2n^{2/3}]$ randomly by choosing each $l \in [n^{2/3}, 2n^{2/3}]$ independently with probability:

$$P(l \in B) = \frac{K \log n}{n^{2/3}}$$

where K will be chosen later. Then, for sufficiently large c, we have that $P(|B| \geq c \log n) < \frac{1}{n}$.

Consider $J = \lfloor \frac{1}{4} \log n \rfloor$ and call $j \in [0, J]$ *good* if:

$$|[2n^{2/3}, n/2^j] \setminus (P + B)| < \frac{\epsilon n}{2^{j+1}}.$$

If j is *good* for all $j \in J$, using that fact that $\frac{2n^{2/3}}{n} < \frac{\epsilon}{2}$ for sufficiently large n, we'll have the required set B. As we want this to hold along with the upper bound on $|B|$ we proved earlier (that fails with probability $< \frac{1}{n}$), we'll be done if we can show that $P(j \text{ is } \textit{good} \text{ for all } j \in J) \geq \frac{1}{n}$

Fix j. Then, for any $m \in [2n^{2/3}, n/2^j]$,

$$\begin{aligned}
P(m \notin P + B) &= P(m - p \notin B, \forall p \in P) \\
&= \prod_{p \in P \cap (m - [n^{2/3}, 2n^{2/3}])} P(m - p \notin B) \\
&= \prod_{p \in P \cap (m - [n^{2/3}, 2n^{2/3}])} \left(1 - \frac{K \log n}{n^{2/3}}\right) \\
&\leq e^{-|P \cap (m - [n^{2/3}, 2n^{2/3}])| \frac{K \log n}{n^{2/3}}} \\
&= e^{-\Omega(K)}
\end{aligned}$$

where we have used the independence of probabilities and the fact that $1-a \leq e^{-a}$. Taking sum over all m,

$$E[|[2n^{2/3}, n/2^j] \setminus (P+B)|] \leq \frac{n}{2^j} e^{-\Omega(K)}.$$

Choosing K large enough and using Theorem 2.1, we get that $P(j \text{ is } \textit{good}) \geq \frac{1}{2}$.

For the final step, as for each j, the event j is *good* is a monotone increasing random variable, by Corollary 2.8, the probability of all j being *good* is at least 2^{-J-1} which is bounded below by $\frac{1}{n}$ for large n as $J = \frac{\log n}{4} + O(1)$. With this, we complete the proof of the lemma. □

Now, we return to the proof of the theorem. Taking the union of all the sets B constructed in the lemma for a fixed $n, n^{4/3}, n^{16/9}, \cdots$, we trivially see that the union (call it U) is a $(1-\epsilon)$-complementary basis of P. Hence, we only need to check the cardinality of U. But that follows from the elementary fact that:

$$|U \cap [1,k]| \leq \log k + \log(k^{3/4}) + \log(k^{9/16}) + \cdots \leq \log k \sum_{i=0}^{\infty} \left(\frac{3}{4}\right)^i = O(\log k)$$

□

With the same notation as above, an asymptotic basis of A of order k is a set X such that $\underline{\sigma}(A + kX) = 1$. Using the same construction as in the previous theorem and interlacing bases for the cases $\epsilon = \frac{1}{2}, \frac{1}{3}, \cdots$ at suitable intervals, we have the following corollary:

Corollary 3.9. *For any sequence $\omega(n)$ going to infinity, there exists an asymptotic basis X of order 1 for P such that $|X \cap [1,n]| \leq \omega(n) \log n$.*

3.3 Miscellaneous

Here, first we see an application in additive combinatorics. Given an additive group A, a subset $B \subseteq A$ is said to be sum-free if there do not exist $x, y, z \in B$ such that $x + y = z$.

Problem 3.10. *Let A be a finite set of non-zero integers. Then, there exists a sum-free subset B of A such that $|B| > |A|/3$.*

Proof. Choose a prime number $p = 3k + 2$ large enough such that $A \subseteq [-p/3, p/3]$. Then, A can be looked upon as a subset of $\mathbb{Z}_p$. In fact, a subset B of A is sum-free in $\mathbb{Z}$ if and only if it is sum-free in $\mathbb{Z}_p$. Now, for a randomly chosen $x \in \mathbb{Z}_p \setminus \{0\}$, construct:

$$B := \{a \in A : x^{-1}a \in \{k+1, k+2, \cdots 2k+1\}\}$$

As $[k+1, 2k+1]$ is a sum-free subset of $\mathbb{Z}_p$, B is a sum-free subset of A. Now,

$$E[|B|] = \sum_{a \in A} P(a \in B) = \sum_{a \in A} P(x^{-1}a \in [k+1, 2k+1]).$$

For any $a \in A$, $a \neq 0$, and so, $x^{-1}a$ is randomly distributed in $\mathbb{Z}_p \setminus \{0\}$. Thus, $P(x^{-1}a \in [k+1, 2k+1]) = \frac{k+1}{3k+1} > \frac{1}{3}$. Hence, $E[|B|] > \frac{|A|}{3}$. Therefore, with positive probabilty, $|B| > |A|/3$, and so for some choice of x, we have the required sum-free subset of A. □

Next, we prove a deep result in analytic number theory which essentially shows that for large n, the number of distinct prime divisors of n is close to $\log \log n$. For a positive integer n, let $\nu(n)$ denote the number of distinct prime divisors of n.

Theorem 3.11. *Let $\omega(n)$ be a sequence tending to ∞. Then,*

$$|x \in [1, n] : |\nu(x) - \log \log x| > \omega(n)\sqrt{\log \log n}| = o(n)$$

Proof. For a fixed n,

$$\begin{aligned} |x \in [1, n] : |\nu(x) - \log \log x| &> \omega(n)\sqrt{\log \log n}| \\ &= nP(|\nu(x) - \log \log x| > \omega(n)\sqrt{\log \log n}), \end{aligned}$$

where the probability is taken over a random choice of x in $[1, n]$. Hence, we only need to show that for a random x, the above probability is $o(1)$. Now, define:

$$B = \{p \leq n^{1/10} : p|x\}$$

where p is a prime. Then, as x can have at most 10 prime factors that are greater than $n^{1/10}$, we have $|B| \leq \nu(x) \leq |B| + 10$. Thus, as $\omega(n) \to \infty$, it suffices to prove that

$$P(||B| - \log\log x| > \omega(n)\sqrt{\log\log n}) = o(1).$$

We can replace $\log\log x$ here by $\log\log n$ as $\log\log x = \log\log n + O(1)$ with probability $1 - o(1)$. In order to prove the above relation, we'll use Theorem 2.3. The result follows directly if we can show:

$$E[|B|], Var(|B|) = \log\log n + O(1)$$

First, we compute the expectation:

$$\begin{aligned} E[|B|] &= \sum_{p \leq n^{1/10}} P(p|x) \\ &= \sum_{p \leq n^{1/10}} \Big(\frac{1}{p} + O\Big(\frac{1}{n}\Big)\Big) \\ &= \sum_{p \leq n^{1/10}} \frac{1}{p} + n^{1/10} O\Big(\frac{1}{n}\Big) = \sum_{p \leq n^{1/10}} \frac{1}{p} + O(n^{-9/10}) \end{aligned}$$

Next, we compute the variance:

$$\begin{aligned} Var(|B|) &= \sum_{p \leq n^{1/10}} (E[I(p|x)^2] - E[I(p|x)]^2) + \sum_{p,q \leq n^{1/10}, p \neq q} Cov(I(p|x), I(q|x)) \\ &= \sum_{p \leq n^{1/10}} \Big(\frac{1}{p} - \frac{1}{p^2} + O\Big(\frac{1}{n}\Big)\Big) + \\ &\quad \sum_{p,q \leq n^{1/10}, p \neq q} \left(\frac{1}{pq} + O\Big(\frac{1}{n}\Big) - \Big(\frac{1}{p} + O\Big(\frac{1}{n}\Big)\Big)\Big(\frac{1}{q} + \Big(\frac{1}{n}\Big)\Big)\right) \\ &= \sum_{p \leq n^{1/10}} \Big(\frac{1}{p} - \frac{1}{p^2}\Big) + O(n^{-9/10}) + \sum_{p,q \leq n^{1/10}, p \neq q} O\Big(\frac{1}{n}\Big) \\ &= \sum_{p \leq n^{1/10}} \Big(\frac{1}{p} - \frac{1}{p^2}\Big) + O(n^{-8/10}) \end{aligned}$$

Now, our claim follows using the fact that $\sum_{p=1}^{\infty} \frac{1}{p^2}$ converges and $\sum_{p \leq n} \frac{1}{p} = O(\log \log n)$. The proofs of these facts can be referred to from any standard analytic number theory text (say [2]). $\square$

Now, we deal with a purely combinatorial problem of colouring. Suppose we wish to colour $\mathbb{R}$ with colours from $\{1, 2, \ldots, k\}$. A subset $T \subseteq \mathbb{R}$ is said to be colourful if it contains all the k colours.

Problem 3.12. *Let m and k be positive integers satisfying:*

$$e(m(m-1)+1)k\Big(1-\frac{1}{k}\Big)^m \leq 1.$$

Then, for any finite subset S of $\mathbb{R}$ with $|S| = m$ and any set $X \subseteq \mathbb{R}$, there exists a k-colouring of $\mathbb{R}$ such that the translates $x + S$ of S are colourful for every $x \in X$.

Proof. First, suppose X is finite. Then, $X + S$ is also finite. Colour every element in $X + S$ randomly and independently from the colours $[1, k]$ with probability $1/k$ for each colour. For any $x \in X$, let A_x be the event that $x+S$ isn't colourful. This happens when at least one of the k colours is missing from $x + S$. Hence, we have the trivial bound:

$$P(A_x) \leq k\Big(1-\frac{1}{k}\Big)^m.$$

Next, we note that A_x and $A_{x'}$ are independent events if $x+S$ and $x'+S$ are disjoint. Also, $x+S$ and $x'+S$ intersect non-trivially if and only if there are elements $s_1, s_2 \in S$ such that $x + s_1 = x' + s_2$. Therefore, for a given x, A_x is dependent on at most $m(m-1)$ (number of ordered pairs in S^2 with unequal elements) other events. Then, putting $p = k\Big(1-\frac{1}{k}\Big)^m$ and $d = m(m-1)$, we're done using Corollary 2.10.

Now, we consider the case when X is arbitrary. The set of all possible colourings of $\mathbb{R}$ can be identified with $[1, k]^{\mathbb{R}}$, which is a compact space in the product topology by Tychonoff's Theorem. For any $x \in \mathbb{R}$, denote K_x to be the subset of this compact space, that corresponds to colourings where $x + S$ is colourful. When looked upon as a subset of $[1, k]^{\mathbb{R}}$, K_x is a product

of finite sets all but finitely many of which are equal to $[1, k]$ (as $x + S$ is finite). Hence, K_x is a clopen set. As we have proven the theorem when X is finite, we have basically shown that the intersection of finitely many K_x's is non-empty. Then, by finite intersection property, intersection of all the K_x's in non-empty, giving us the required colouring of $\mathbb{R}$. □

Our final application deals with the concept of antichains. A collection of sets S is called an antichain if for any $A, B \in S$, we have $A \not\subseteq B$ unless $A = B$.

Theorem 3.13. *(LYM inequality) Let $\mathcal{A}$ be an antichain consisting of subsets of a finite set X. Then,*

$$\sum_{A \in \mathcal{A}} \frac{1}{\binom{|X|}{|A|}} \leq 1$$

Proof. Let $\phi : X \to \{1, 2, \ldots, |X|\}$ be a random bijection chosen uniformly from among all such bijections. Then,

$$P(\phi(A) = \{1, 2, \cdots, |A|\}) = \frac{|A|!(|X| - |A|)!}{|X|!} = \frac{1}{\binom{|X|}{|A|}}.$$

For any two unequal $A, B \in \mathcal{A}$, as neither contains the other, the events $\phi(A) = \{1, 2, \cdots, |A|\}$ and $\phi(B) = \{1, 2, \cdots, |B|\}$ are disjoint. Therefore, summing over the probabilities of all these disjoint events, the final result should be bounded above by 1, which was exactly what we wanted to prove. □

References

[1] Terence Tao, Van H. Vu, Additive Combinatorics, Cambridge Studies in Advanced Mathematics 105

[2] Paul Pollack, Not Always Buried Deep, Selections from Analytic and Combinatorial Number Theory, 2003